Sign Robustness to Imputation

Empirical researchers often replace missing or interval-observed values using an imputation rule before estimating a regression. Suppose the resulting coefficient is positive. Could another completion of the same incomplete data produce a negative coefficient? Researchers may care more about this qualitative robustness question than about estimating the entire identified set of regression coefficients.

This project proposes a direct sign-reversal test and applies it systematically to empirical papers published in leading economics journals. The central idea is that the sign of a regression coefficient can be substantially easier to analyze than its magnitude.

1. Main idea

Consider a linear regression of Y on a scalar regressor of interest X and controls Z, including an intercept. Initially, assume Y and Z are observed, while $X_i \in [X_{i,L}, X_{i,U}]$. Denote an admissible completion by $X(\mathcal{I})$. By the Frisch–Waugh–Lovell theorem,

$$\hat{\beta}(\mathcal{I}) = \frac{X(\mathcal{I})' M_Z Y}{X(\mathcal{I})' M_Z X(\mathcal{I})}, \qquad M_Z = I - Z(Z'Z)^{-1}Z'. \tag{1}$$

For every completion with a nonzero residualized regressor, the denominator is positive. Writing $r = M_Z Y$, the sign therefore depends only on $X(\mathcal{I})'r$, which is *linear* in the imputed regressor.

The smallest feasible numerator is

$$\underline{S}_n = \min_{X_i \in [X_{i,L}, X_{i,U}]} \sum_{i=1}^{n} X_i r_i = \sum_{i=1}^{n} \min\{X_{i,L} r_i,\ X_{i,U} r_i\}. \tag{2}$$

If the reported coefficient is positive, a strictly negative coefficient is feasible if and only if $\underline{S}_n < 0$. This is an exact sample statement: no search over regression coefficients or nonlinear optimization is required. A parallel upper bound $\overline{S}_n$ addresses negative reported coefficients.

2. Missing outcomes and other imputed controls

Missing outcome. If X and Z are fixed but $Y_i \in [Y_{i,L}, Y_{i,U}]$, let $t = M_Z X$. The same argument gives

$$\underline{S}_n^Y = \sum_{i=1}^{n} \min\{t_i Y_{i,L},\ t_i Y_{i,U}\}. \tag{3}$$

Thus, the simple test applies to either an incomplete regressor of interest or an incomplete outcome.

Other imputed controls. In practice, several control variables may also be missing. Let Z^0 be the matrix obtained using the researcher's original imputation rules. Holding Z^0 fixed, the first formula applies unchanged with $r^0 = M_{Z^0} Y$. The test is then *conditional on the imputations of the other controls*: it asks whether changing the imputation of X alone can reverse the reported result. It is sharp for that question, but does not establish robustness to simultaneously changing every imputed control. One can also repeat the test for several prespecified nuisance-imputation rules.

Both X and Y incomplete. The product $X_i Y_i$ has its extrema at the four corners of the observed rectangle. With an intercept or other controls, however, the relevant expression is $X'M_Z Y$, not $\sum_i X_i Y_i$. Its optimization generally couples observations; evaluating four corners separately for each observation is not sufficient. Jointly incomplete outcomes and regressors are therefore a useful extension, but need not admit the same closed-form sum.

3. Population inference

The sample test answers whether a different completion can reverse the coefficient in the observed sample. Population inference concerns the sign of the best-linear-prediction (BLP) coefficient across distributions consistent with the incomplete observations. Let

$$\pi_0 = E[ZZ']^{-1}E[ZY], \qquad R = Y - Z'\pi_0. \tag{4}$$

When Y, Z are observed and X is interval-valued, the sharp lower bound for the population regression numerator is

$$\underline{S} = \inf_{X \in \mathrm{Sel}([X_L, X_U])} E[XR] = E[\min\{X_L R, X_U R\}]\,. \tag{5}$$

Assuming all relevant BLP denominators are positive, $\underline{S} > 0$ means that every admissible population BLP coefficient is positive, while $\underline{S} < 0$ means that a negative coefficient is feasible. To *certify positive sign robustness*, test

$$H_0 : \underline{S} \leq 0 \qquad \text{against} \qquad H_1 : \underline{S} > 0. \tag{6}$$

A confidence interval for $\underline{S}$ can also provide evidence of a possible reversal if it lies entirely below zero. Estimation uses the sample average of $\min\{X_{i,L}\hat{R}_i, X_{i,U}\hat{R}_i\}$. Inference must account for estimated residualization coefficients, any data-dependent nuisance imputations, and potential non-smoothness at zero. Joint inference for lower and upper bounds can avoid a separate correction for selecting which sign to examine from the reported estimate.

4. Meta-study of published research

The empirical contribution would be a systematic reanalysis of empirical articles in the *American Economic Review*, *Econometrica*, *Journal of Political Economy*, *Quarterly Journal of Economics*, and *Review of Economic Studies* over the ten complete publication years 2016–2025.

The study would screen articles for missing or interval-observed outcomes and covariates, reproduce the main reported specifications where replication materials permit, and apply the sign test to eligible results. We would report how often signs are robust in the sample and how often robustness or reversal can be established statistically. The study would also document the fraction of papers for which the test cannot be applied and why (e.g., unavailable pre-imputation data, unsupported estimators, or uninformative feasible ranges). Cases requiring additional support assumptions would be reported separately from those with observed interval restrictions.

The goal is not merely to give examples, but to quantify how often published qualitative findings depend on imputation choices. A preliminary pilot on a smaller set of papers would establish the feasibility of reconstructing the original missingness information.

5. Relation to existing work and contribution

Beresteanu and Molinari (2008) and Beresteanu, Molchanov, and Molinari (2011, BMM) develop random-set methods for partial identification; BMM explicitly treats BLP with interval outcomes and covariates. We do not replace their identification framework. Rather, we exploit the fact that a *sign* question can reduce to a single scalar bound, avoiding computation of the full BLP coefficient region. The project also differs from Fankhauser and Peled (2026), which imposes conditional moment restrictions: the proposed sign test concerns the BLP and does not require those additional model restrictions.

Ober-Reynolds (2026) studies missing-data robustness using a breakdown point for deviations between distributions of complete and incomplete observations. Palomba (2026, working paper) studies robustness to distributional deviations indexed by a sensitivity budget. Our focus is different: we hold the feasible sets implied by incomplete observations fixed and ask whether admissible imputations can reverse a reported regression sign. Brodeur et al. (2026) illustrate the value of large-scale robustness replications, but do not study this specific imputation question.

The proposed contribution is a computationally cheap, directly interpretable sign-robustness test, associated inference, and systematic evidence on the role of imputation in leading empirical research. Its scope is deliberately narrower than general partial identification: conditioning on nuisance imputations yields a useful applied diagnostic, not a guarantee of robustness to all jointly varying missing values.

References

Beresteanu, A. and F. Molinari (2008). "Asymptotic Properties for a Class of Partially Identified Models." *Econometrica* 76(4), 763–814.

Beresteanu, A., I. Molchanov, and F. Molinari (2011). "Sharp Identification Regions in Models With Convex Moment Predictions." *Econometrica* 79(6), 1785–1821.

Brodeur, A., D. Mikola, N. Cook, et al. (2026). "Reproducibility and Robustness of Economics and Political Science Research." *Nature* 652, 151–156.

Fankhauser, M. and G. Peled (2026). "Sharp Identification for Regressions with Interval-Observed and Missing Covariates." Preliminary draft.

Ober-Reynolds, D. (2026). "Robustness to Missing Data: Breakdown Point Analysis." *Journal of Econometrics* 253, article 106151.

Palomba, F. (2026). "Sensitivity Analysis in Population Shares." Working paper; description and software at `https://filippopalomba.github.io/`.